

Kai Wu

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RESEARCH INTERESTS

Differential Geometry, Geometric Analysis, Partial Differential Equations, Minimal Surfaces

EDUCATION

Peking University, Ph.D. in Mathematics 2025 – Present
Supervisor: Prof. Gang Tian

University of Science and Technology of China, B.Sc. in Mathematics 2021 – 2025
Supervisor: Prof. Yong Wei GPA: 3.93/4.3 (Top 5%)

RESEARCH EXPERIENCE

Min-Max Theory and Willmore Conjecture 2025
Bachelor's Thesis, supervised by Prof. Yong Wei and Prof. Gang Tian

- Studied the proof of the Willmore conjecture by Marques and Neves and learned min-max theory for minimal surfaces in Riemannian manifolds.

Topics on Minimal Surfaces 2024 – 2025
Undergraduate Innovation Training Program, supervised by Prof. Yong Wei

- Learned fundamental properties of minimal surfaces, including Bernstein's theorem in low dimensions and counterexamples in higher dimensions.

RELEVANT COURSEWORK

Riemannian Geometry; Algebraic Topology; Second-Order Elliptic PDEs; Geometric Analysis; Morse Theory (self-study); Geometric Measure Theory (self-study)

AWARDS AND HONORS

- XXX Scholarship, 2024
- Mathematical Contest in Modeling (MCM), Honorable Mention

TALKS AND PRESENTATIONS

- “Min-Max Construction and Willmore Conjecture,” Academic Symposium for Students of the Honors Program in Basic Sciences, USTC, June 2025

TEACHING EXPERIENCE

- Advanced Mathematics (B) II, Peking University, Spring 2026
<https://kaiwu2003.github.io/calc-spring-2026/>
- Linear Algebra (B), Peking University, Fall 2025
- Linear Algebra (B), USTC, Autumn 2024
- Linear Algebra (B), USTC, Spring 2024